



May 2025

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1-

8. Which of the following keywords is used to inherit a class in Java?
A) extends B) implements
C) inherits D) superclass
9. Which of these methods is used to pause the execution of a thread?
A) wait() B) sleep()
C) pause() D) stop()
10. Which JDBC driver is fully written in Java?
A) Type 1 B) Type 2
C) Type 5 D) Type 4
11. Which of the following is the base class for all exceptions in Java?
A) Exception B) Throwable
C) RuntimeException D) Error
12. What will be the output of the following code?
try {
 int a = 10 / 0;
} catch (ArithmeticException e) {
 System.out.println("Division by zero");
}
A) Compilation error B) Runtime exception
C) No output D) Division by zero
13. Which method is used to start a thread in Java?
A) run() B) init()
C) start() D) begin()
14. Which method is used to execute a SQL query in JDBC?
A) executeStatement() B) runQuery()
C) runStatement() D) executeQuery()
15. What is the purpose of the finally block in exception handling?
A) To execute code after the catch block, regardless of exception B) To define another exception type
C) To handle system errors D) To rethrow an exception
16. What will be the output of the following code?
class MyThread extends Thread {
 public void run() {
 System.out.println("Thread is running");
 }
}
public class Test {
 public static void main(String[] args) {
 MyThread t = new MyThread();
 t.start();
 }
}
A) Compilation error B) Thread is running
C) Runtime exception D) No output
17. Which method is used to send the same request and response objects to another servlet?
A) forward() B) next()
C) sendForward() D) sendRedirect()
18. What is the difference between servlets and applets?
i. Servlets execute on the Server; Applets execute on the browser
ii. Servlets have no GUI; Applet has GUI
iii. Servlets creates static web pages; Applets creates dynamic web pages
iv. Servlets can handle only a single request; Applet can handle multiple requests
A) i, ii, iii are correct B) i, ii are correct
C) i, iii are correct D) i, ii, iii, iv are correct

19. Except for the _____ phase, a JSP page is handled exactly like a Servlet
 A) Translation B) Request
 C) Response D) Initialization
20. JSON stands for _____.
 A) JavaScript Object Normalization B) Java Object Notation
 C) JavaScript Object Notation D) JavaScript Object-Oriented Notation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How do you implement run-time polymorphism using Dynamic method dispatch? Explain in detail.	8	L*2	2	1
	b) With the help of a simple Java program differentiate between for-loop and for-each loop	8	L2	1	1
2.	a) Write a short note on the following i. Three OOP principles ii. Java string class	8	L2	1	1
	b) With the help of an example program demonstrate interfaces in Java.	8	L2	2	1
3.	a) With the help of the Java program explain the concept of returning objects.	8	L2	1	1
	b) Create a class with a method that prints "This is parent class" and its subclass with another method that prints "This is child class". Now, create an object for each of the classes and call i. - method of parent class by the object of the parent class ii. - method of child class by object of child class iii. - method of parent class by object of child class	8	L3	2	1
Unit – II					
4.	a) What are the advantages of the Multithreading? With a suitable block diagram explain the life cycle of thread.	8	L2	3	1
	b) Illustrate the use of the Finally block in Java with suitable example codes.	8	L2	3	1
5.	a) What is Exception in Java? Explain the exception-handling fundamentals with suitable examples.	8	L2	3	1
	b) What is Java Database Connectivity? Explain its architecture.	8	L1	4	1
6.	a) What is a JDBC driver? Explain four different types of JDBC drivers in detail.	8	L2	4	1
	b) Write a short note on the following i. Extending threads ii. Thread Priorities	8	L2	3	1
Unit – III					
7.	a) What are servlets? What are its benefits? Explain its architecture.	8	L2	5	1
	b) Explain different types of JSP tags with Examples.	8	L1	5	1
8.	a) Explain the JSP life cycle in detail with the necessary syntax.	8	L2	5	1
	b) What is JSON? Mention its characteristics and uses.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
